Quadratic Witt Classes over $\mathbb{F}_2(t)$: Residue Coordinates, Ramification, and Impartial Realization

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Abstract

Let $K = \mathbb{F}_2(t)$. Its quadratic Witt group contains wild local data that is invisible to the one-bit Arf classification over finite fields. Starting from the characteristic-two Milnor–Scharlau sequence of Aravire and Jacob, we give a canonical global coordinate theorem: a nonsingular Witt class is determined by its unramified coordinate at infinity, its finite odd-pole residues, and its ramified Arf bits, subject only to one parity relation. The key transfer calculation is proved directly from trace duality.

These coordinates admit a faithful additive normal-play realization. Each residue-field coefficient is observed by all trace characters, and each resulting bit is represented by the literal impartial values 0 and $*$. Every class therefore emits a finite sparse labelled family, orthogonal sum becomes coordinatewise disjunctive sum, and equality of outcome families is exactly Witt equivalence. We also compute the finite triangular coordinate law for ramified scalar extension $\pi = u\varpi^e$ and its adjoint action on trace labels.

Two sharp boundaries complete the result. No fixed finite family can be faithful on all refinements of even one binary polar space, and no additive map to a cancellative game-value monoid can retain arbitrary quasilinear radical data. The Lean development constructs the complete presented coordinate-to-game compiler and checks the transfer, ramification, finiteness, and singular-isometry cores; the cited Aravire–Jacob classification is the single external bridge to $W_q(\mathbb{F}_2(t))$.

1 Introduction

Over a finite field of characteristic two, a nonsingular quadratic Witt class is controlled by one Arf bit [3, 5]. That bit has an immediate impartial interpretation: 0 is a previous-player win, $* = \{0 \mid 0\}$ is a next-player win, and disjunctive sum is xor [4]. It is tempting to expect the same one-bit picture over the rational function field $\mathbb{F}_2(t)$. The obstruction is not game-theoretic. It is arithmetic: $\mathbb{F}_2(t)$ is imperfect, and odd negative Laurent powers survive Artin–Schreier reduction at every place.

The correct realization problem is therefore a coordinate problem. One must first identify the global Witt class from its local data, then turn every local coefficient into invariant binary observations, and only then attach games. This order separates a genuine additive realization from the Boolean serialization of one chosen formula for a quadratic form.

The arithmetic input is the characteristic-two Milnor–Scharlau sequence of Aravire and Jacob [2, Theorem 6.2]:

$$0 \longrightarrow W_q(\mathbb{F}_2) \longrightarrow W_q(K) \xrightarrow{\oplus_v \partial_v} \bigoplus_v W_1(K_v) \xrightarrow{\oplus_v s_v^*} W_q(\mathbb{F}_2) \longrightarrow 0.$$

Here $W_1(K_v)$ is a quotient of the Witt group of the completion, not the Witt group of the residue field. Its wild summand is infinite-dimensional, but every individual class has finite place and pole support. We show that the endpoint transfer reads only the ramified Arf bit. The infinite place then supplies a canonical retraction of the constant-field inclusion, giving an explicit global splitting rather than an abstract choice of basis.

The realization built from this splitting is deliberately sparse. Its label universe is infinite because the set of places and pole orders is infinite; the family emitted by any one class is finite. This distinction leads to the first optimality theorem: finite per instance is possible, while one fixed finite directory for all refinements is not. A second, independent obstruction appears for singular forms, whose orthogonal-sum monoid is already noncancellative in dimension one.

Theorem A. *Fix $K = \mathbb{F}_2(t)$, use the monic irreducible polynomial $p(t)$ as uniformizer at the finite place p, and use t^{-1} at infinity.*

1. *There is a canonical additive isomorphism*

$$W_q(K) \cong \mathbb{F}_2 \oplus \left\{ ((\psi_v, \epsilon_v))_v : \textit{finite support and } \sum_v \epsilon_v = 0 \right\},$$

   where ψ_v is a finite odd-pole residue over κ_v and ϵ_v is the ramified Arf bit.

2. *Trace characters give a basis-free finite sparse compiler into labelled copies of the impartial values 0 and $*$. It is additive under orthogonal sum, and two compiled outcome families agree exactly when their source classes are Witt equivalent.*

3. *Under $k((\pi)) \hookrightarrow \ell((\varpi))$ with $\pi = u\varpi^e$, restriction is an explicit finite triangular transformation of the odd-pole coefficients. It is functorial in towers and has an explicit contravariant adjoint on trace labels.*

4. *No fixed finite refinement-independent P/N family is faithful on all refinements of even one nonsingular binary polar space. Likewise, no isometry-invariant orthogonal-sum homomorphism into a cancellative game-value monoid can distinguish every quasilinear radical restriction.*

The exact sequence and local decomposition are cited inputs. The finite-field transfer calculation, canonical splitting, trace-bit compiler, ramified coordinate transformation, and both optimality theorems are proved here. The formalization follows the same boundary: it treats the cited classification as one explicit equivalence and constructs every subsequent map and game internally.

2 Local residues and global reciprocity

2.1 The local normal form

Let v range over the monic irreducible polynomials $p \in \mathbb{F}_2[t]$ and the infinite place. Define

$$\pi_p = p(t), \qquad \pi_\infty = t^{-1}.$$

Write K_v for the completion and κ_v for its residue field. Thus

$$\kappa_p = \mathbb{F}_2[t]/(p), \qquad \kappa_\infty = \mathbb{F}_2.$$

Every κ_v is finite and therefore perfect. The local theorem of Aravire and Jacob specializes to [2, Theorem 1.3 and Corollary 1.7]

$$W_q(K_v) \cong W_q(\kappa_v) \oplus R_v \oplus \langle \pi_v \rangle W_q(\kappa_v),$$

where

$$R_v = \bigoplus_{\substack{n\geq 1\\ n \text{ odd}}} \kappa_v \pi_v^{-n}.$$

The first summand is unramified. Quotienting it out gives

$$W_1(K_v) := \operatorname{coker}\bigl(W_q(\kappa_v) \longrightarrow W_q(K_v)\bigr) \cong R_v \oplus \mathbb{F}_2.$$

For a global class α, write its localization as

$$\operatorname{loc}_v(\alpha) = (\phi_{0,v}, \psi_v, \phi_{1,v}), \qquad \psi_v = \sum_{n \text{ odd}} c_{v,n}\pi_v^{-n}.$$

Then

$$\partial_v \alpha = (\psi_v, \phi_{1,v}).$$

Only finitely many places occur for one global class, and each ψ_v is a finite sum. The uniformizer is part of this description: changing it changes the displayed odd-pole coordinates even when the underlying class is fixed.

2.2 The endpoint transfer

The local decomposition has many wild coordinates, but the final arrow in the global exact sequence is much simpler. Its identification is the step that turns reciprocity into one parity equation. The proof uses only the nondegenerate trace pairing of a finite-field extension [6].

Lemma 2.1 (Transfer preserves the Arf generator)**.** *Let $E/\mathbb{F}_2$ be finite and let $s : E \to \mathbb{F}_2$ be a nonzero linear functional. Under the Arf identifications*

$$W_q(E) \cong \mathbb{F}_2 \cong W_q(\mathbb{F}_2),$$

the Scharlau transfer $s_* : W_q(E) \to W_q(\mathbb{F}_2)$ is the identity.

Proof. The trace pairing is nondegenerate, so $s(z) = \operatorname{Tr}(cz)$ for a unique $c \neq 0$. Since E is perfect, choose $d \in E$ with $d^2 = c$. Let $a \in E$ satisfy $\operatorname{Tr}(a) = 1$; the plane $[1, a]$ represents the nonzero class in $W_q(E)$. Its transfer along s is

$$s(x^2 + xy + ay^2) = \operatorname{Tr}(cx^2 + cxy + cay^2).$$

After the simultaneous change of variables $X = dx$, $Y = dy$, this is

$$T_a(X, Y) = \operatorname{Tr}(X^2 + XY + aY^2).$$

Choose a basis (e_i) of $E/\mathbb{F}_2$ and its trace-dual basis (f_i). The vectors $(e_i, 0), (0, f_i)$ form a symplectic basis for the polar form of T_a. Therefore

$$\begin{aligned}\operatorname{Arf}(T_a) &= \sum_i \operatorname{Tr}(e_i^2)\operatorname{Tr}(af_i^2)\\ &= \operatorname{Tr}\left(a \sum_i \operatorname{Tr}(e_i) f_i^2\right).\end{aligned}$$

Trace duality applied to $1 \in E$ gives

$$1 = \sum_i \mathrm{Tr}(e_i) f_i.$$

Squaring in characteristic two gives $1 = \sum_i \mathrm{Tr}(e_i) f_i^2$. Hence

$$\mathrm{Arf}(T_a) = \mathrm{Tr}(a) = 1.$$

The nonzero class is therefore carried to the nonzero class. $\square$

In the Aravire–Jacob construction, s_v^* kills R_v and applies ordinary Scharlau transfer to the ramified residue form. Lemma 2.1 therefore gives

$$s_v^*(\psi_v, \phi_{1,v}) = \phi_{1,v}.$$

Thus wild coefficients are locally essential but globally unconstrained; the only reciprocity relation is parity of the ramified bits.

2.3 The canonical global splitting

The residue tuple determines a global class only modulo the constant-field image. The unramified coordinate at the fixed infinite place supplies the missing coordinate canonically; no vector-space complement is chosen.

Theorem 2.2 (Infinity plus second residues)**.** *Define*

$$\Theta(\alpha) = \left(\phi_{0,\infty}(\alpha), ((\psi_v(\alpha), \phi_{1,v}(\alpha)))_v\right).$$

Then Θ is an additive isomorphism

$$W_q(\mathbb{F}_2(t)) \xrightarrow{\sim} \mathbb{F}_2 \oplus \left\{ ((\psi_v, \epsilon_v))_v : \textit{finite support and } \sum_v \epsilon_v = 0 \right\}.$$

Proof. The transfer formula above identifies the kernel of the final arrow in the Milnor–Scharlau sequence with the displayed parity-zero residue tuples. The unramified coordinate $\phi_{0,\infty}$ is additive. On a constant-field class $a \in W_q(\mathbb{F}_2)$, localization at infinity has coordinates $(a, 0, 0)$, so $\phi_{0,\infty}$ restricts to the identity on constants.

Suppose $\Theta(\alpha) = 0$. All second residues vanish, so exactness gives α in the image of $W_q(\mathbb{F}_2)$. Its infinite unramified coordinate is that constant class, and hence the first coordinate makes it zero. This proves injectivity.

Conversely, let r be a finite residue tuple of parity zero. Exactness gives $\beta \in W_q(K)$ with residue r. Given a prescribed constant coordinate $a \in \mathbb{F}_2$, set

$$\alpha = \beta + \big(a - \phi_{0,\infty}(\beta)\big),$$

where the parenthesized term is included as a constant class. It changes no second residue and changes the first coordinate to a. Thus Θ is surjective. Injectivity makes the lift unique. $\square$

Remark 2.3. Exactness alone gives a noncanonical splitting because these groups are $\mathbb{F}_2$-vector spaces. The theorem is stronger: the first local coordinate at the fixed infinite place is an explicit retraction, so no basis extension or choice of residue lifts appears in Θ.

3 The sparse impartial compiler

Theorem 2.2 reduces the realization problem to binary observation of finite-field coefficients. A power-basis expansion would do this finitely but noncanonically. Trace duality gives a redundant, basis-free replacement.

Definition 3.1 (Trace-bit presentation)**.** For $z \in \kappa_v$, define

$$b_{v,n,z}(\alpha) = \mathrm{Tr}_{\kappa_v/\mathbb{F}_2}(zc_{v,n}).$$

Also put

$$b_{\mathrm{c}}(\alpha) = \phi_{0,\infty}(\alpha), \qquad b_{v,\mathrm{r}}(\alpha) = \phi_{1,v}(\alpha).$$

Each bit is additive in α. Nondegeneracy of the trace pairing says that $c_{v,n} = 0$ exactly when every $b_{v,n,z}$ vanishes. The complete labelled bit function is therefore faithful by Theorem 2.2. It is sparse: one class has finitely many active places and pole orders, and each residue field is finite, so only finitely many labels can carry a nonzero bit.

Let 0 denote the optionless impartial game and let $* = \{0 \mid 0\}$. These are the nimbers 0 and 1, so $* + * = 0$ under disjunctive sum [4]. Equivalently, each bit is a literal empty-core arena whose terminal one-move tail is present exactly at charge one.

Theorem 3.2 (Sparse impartial compiler)**.** *For every nonsingular quadratic Witt class α over $\mathbb{F}_2(t)$, replace each zero trace, constant, or ramified bit by the zero game and each one bit by $*$. Omit zero entries from the resulting labelled family. Then:*

1. *the family is finite and a coordinate is a P-position exactly when its bit is zero;*
2. *orthogonal sum becomes coordinatewise disjunctive sum;*
3. *hyperbolic classes map to the zero family; and*
4. *two families have the same outcome vector exactly when their source forms are Witt equivalent.*

Proof. Finiteness and faithfulness were proved above. The map $\mathbb{F}_2 \to \{0, *\}$ is an additive injection because $* + * = 0$. Applying it to every labelled bit proves the orthogonal-sum identity. Theorem 2.2 then identifies equality of all bits with equality in $W_q(K)$; the zero class consists precisely of hyperbolic forms. $\square$

The order of construction is essential. Every emitted bit is a homomorphism on the Witt group, defined only after passage through the residue theorem; it is therefore invariant under isometry and hyperbolic stabilization. The compiler does not claim to discover a residue by local play. It computes the canonical class coordinates and realizes their values by games. The precise cost of requiring a refinement-blind local arena is addressed in Section 5.1.

4 Ramified scalar extension

The compiler is functorial only after the coordinate law under scalar extension is known. We now compute that law locally. Let

$$K = k((\pi)) \hookrightarrow L = \ell((\varpi)), \qquad \pi = u\varpi^e,$$

where k and ℓ are finite fields, $f = [\ell : k]$, and $u \in \ell[[\varpi]]^\times$. Write $\bar{e}, \bar{f} \in \mathbb{F}_2$ for the parities of e, f. For every positive odd n, expand

$$u^{-n} = \sum_{r\geq 0} b_{n,r}\varpi^r.$$

Inverse Frobenius in the perfect field ℓ is denoted by $x \mapsto x^{2^{-s}}$.

4.1 Finite Artin–Schreier normalization

The elementary Artin–Schreier reduction is finite and triangular. If $A = \sum_j A_j \varpi^j$ has finite principal part, then its canonical odd-pole coordinates are

$$\begin{aligned}
\epsilon(A) &= \mathrm{Tr}_{\ell/\mathbb{F}_2}(A_0),\\
\rho_d(A) &= \sum_{s\geq 0} A_{-2^s d}^{2^{-s}} \qquad (d > 0 \text{ odd}).
\end{aligned}$$

Indeed, adding

$$\wp(c\varpi^{-m}) = c^2\varpi^{-2m} + c\varpi^{-m}$$

kills an even pole and changes only the strictly smaller pole m. Descending cancellation gives the displayed sum. Positive powers lie in $\wp(\varpi\ell[[\varpi]])$, while an odd leading pole cannot occur in an Artin–Schreier image. The constant coefficient is unchanged by pole cancellation; any new constant below comes from the unit jet before reduction.

For a wild input $\psi = \sum_{n>0 \text{ odd}} c_n\pi^{-n}$, define

$$\begin{aligned}
\eta_{e,u}(\psi) &= \sum_n \mathrm{Tr}_{\ell/\mathbb{F}_2}\big(\iota(c_n)b_{n,en}\big),\\
C_d(\psi) &= \sum_n \sum_{\substack{s\geq 0\\ 2^s d\leq en}} \big(\iota(c_n)b_{n,en-2^s d}\big)^{2^{-s}},\\
T_{e,u}(\psi) &= \sum_{d>0 \text{ odd}} C_d(\psi)\varpi^{-d}.
\end{aligned}$$

All sums are finite, and $C_d = 0$ for $d > e \max \mathrm{supp}(\psi)$.

4.2 The restriction law

Theorem 4.1 (Ramified restriction law)**.** *In the local coordinates*

$$W_q(K) \cong \mathbb{F}_2 \oplus R_\pi \oplus \mathbb{F}_2,$$

scalar restriction to L is

$$(\phi_0, \psi, \phi_1) \longmapsto \left(\bar{f}\phi_0 + \eta_{e,u}(\psi) + (1-\bar{e})\bar{f}\phi_1, T_{e,u}(\psi), \bar{e}\bar{f}\phi_1\right).$$

These transformations compose in towers.

Proof. Substitution sends $c_n\pi^{-n}$ to

$$\iota(c_n)u^{-n}\varpi^{-en}.$$

Its coefficient at exponent $-2^s d$ is $\iota(c_n)b_{n,en-2^s d}$, so the preceding odd-pole normal form gives exactly C_d. Its coefficient at exponent zero is $\iota(c_n)b_{n,en}$, giving $\eta_{e,u}$.

Choose $a \in k$ with $\mathrm{Tr}_{k/\mathbb{F}_2}(a) = 1$. The unramified generator $[1, a]$ restricts to $\bar{f}$ times the unramified generator because $\mathrm{Tr}_{\ell/\mathbb{F}_2}(\iota(a)) = \bar{f}$. For the ramified generator, put $A = u\varpi^e$ and use

$$\langle \pi \rangle [1, a] = [A, \iota(a)/A].$$

Write $A = A_{\mathrm{ev}} + A_{\mathrm{odd}}$ by parity of the ϖ-exponent. The local block decomposition is

$$[A, \iota(a)/A] = [1, A_{\mathrm{ev}}\iota(a)/A] \perp \langle \varpi \rangle [1, A_{\mathrm{odd}}\iota(a)/A].$$

Both additive slots are integral and their constant terms are respectively $\iota(a), 0$ when e is even and $0, \iota(a)$ when e is odd. Their positive terms are Artin–Schreier trivial. Thus the ramified bit becomes unramified for even e, remains ramified for odd e, and is multiplied by $\bar{f}$; no wild term remains. This proves the formula.

For a tower $K \subset L \subset M$, both iterated transformation and direct transformation represent actual scalar restriction followed by the unique odd-pole normal form. Uniqueness makes them equal. □

4.3 Trace-label adjunction

The trace labels admit an explicit contravariant transformation. For $\lambda \in \mathbb{F}_2$ and a finite family $z_d \in \ell$, set

$$Z_n = \mathrm{Tr}_{\ell/k}\left(\lambda b_{n,en} + \sum_{\substack{d,s \\ 2^s d \le en}} z_d^{2^s} b_{n,en-2^s d} \right).$$

Frobenius invariance and trace transitivity give

$$\lambda \eta_{e,u}(\psi) + \sum_d \mathrm{Tr}_{\ell/\mathbb{F}_2}(z_d C_d(\psi)) = \sum_n \mathrm{Tr}_{k/\mathbb{F}_2}(Z_n c_n).$$

Hence base change is a basis-free relabelling of the trace-bit games, and the adjoints compose contravariantly in towers.

There is one necessary qualification concerning the word "transfer." The Aravire–Jacob endpoint map $W_1(K) \to W_q(k)$ kills the wild summand and reads the ramified bit. Theorem 4.1 therefore gives the usual ramification-weighted diagram

$$\sigma_L \circ \mathrm{res}_{L/K} = \bar{e}\ \mathrm{res}_{\ell/k} \circ \sigma_K.$$

An arbitrary Scharlau transfer does not descend to W_1. Even for $L = K$, the functional $s(x) = \pi x$ sends an unramified generator to its nonzero $\langle \pi \rangle$ multiple. Thus a stronger transfer claim must specify a normalized functional; the endpoint diagram and trace-character adjunction above are the canonical statements.

5 Optimality and realization boundaries

The preceding construction is complete for nonsingular Witt classes, but it does not satisfy every stronger realization requirement one might impose. Three distinctions are forced by the algebra: finite support is not the same as a fixed finite label universe, Witt invariance is not pointwise evaluation, and the nonsingular Witt group is not the monoid of all singular quadratic spaces. We make each boundary precise.

5.1 No fixed finite local-access family

The finite compiler of Theorem 3.2 omits its zero coordinates after inspecting the class. A strict Gold-style local realization would ask for more: with the polar form fixed, one finite list of position universes, loadings, and coordinate labels must be independent of the refinement, while only move predicates may query its diagonal. That conjunction is inconsistent over $\mathbb{F}_2(t)$.

Theorem 5.1 (No fixed finite local-access family)**.** *Let $K = \mathbb{F}_2(t)$. No fixed finite P/N family, independent of the quadratic refinement, is faithful on all nonsingular refinements of one fixed binary polar space. Consequently it cannot equal the canonical Witt-coordinate family.*

Proof. On $V = K^2$ fix the polar form $B(e, f) = 1$. For every monic irreducible $P \in \mathbb{F}_2[t]$, consider

$$q_P(x, y) = x^2 + xy + P^{-1}y^2 = [1, P^{-1}].$$

All q_P have polar form B. At the place P, with $\pi_P = P$, the local decomposition is

$$(\phi_0, \psi, \phi_1) = \big(0, \pi_P^{-1}, 0\big).$$

At every other finite place there is no negative Laurent term. The classes $[q_P]$ are therefore pairwise distinct by the global coordinate theorem. There are infinitely many monic irreducibles.

If the fixed family has N roots, its P/N vector lies in $\mathbb{F}_2^N$, a finite set. It cannot distinguish the infinite family $([q_P])_P$. The conclusion is independent of how many diagonal queries a transition makes. □

The obstruction isolates the exact alternatives. A family indexed by every place and pole label can keep refinement-independent boards, but its label universe is infinite, although each class has finite nonzero support. Selecting only that support gives the finite refinement-dependent sparse compiler of Theorem 3.2. Restricting the input to a fixed finite set of places and a bounded conductor instead gives a finite static family. These are coherent relaxations; none satisfies the literal conjunction ruled out by the theorem.

For one requested coordinate, an on-demand local arena is available. Choose a public symplectic frame (u_i, v_i). The class is the orthogonal sum of $[Q(u_i), Q(v_i)]$. If $a = a_{\mathrm{ev}} + a_{\mathrm{odd}}$ is the Laurent-parity split, the explicit per-plane contribution at v is

$$D_v([a, b]) = \big(\epsilon_v(a_{\mathrm{ev}}b), \rho_v(ab), \epsilon_v(a_{\mathrm{odd}}b)\big).$$

Thus the requested coordinate is the xor of elementary constant, ramified, or trace evaluations of these D_v values. A source-only matching FIFO board can query $Q(u_i), Q(v_i)$ when the source pair opens and charge that local bit. The public matching strategy forces the total charge, and a double one-option suspension normalizes the value to exactly 0 or $*$. This uses two local diagonal queries per source transition, but it is an on-demand coordinate arena, not a fixed finite family covering every possible place.

5.2 Pointwise realization versus Witt realization

The distinction between pointwise quadratic realization and Witt realization is structural, not merely an implementation choice.

Proposition 5.2 (Pointwise loadings do not descend)**.** *A Witt-invariant additive family cannot literally reproduce the complete loaded-root family of a pointwise quadratic realization. It can recover only its Witt/Arf shadow.*

Proof. The hyperbolic plane

$$H(x,y) = xy$$

is zero in $W_q(\mathbb{F}_2)$. Every Witt-invariant realization must therefore map H to the zero family. But $H(1,1) = 1$, so any exact pointwise realization has an N-position at the loading $(1,1)$. The pointwise outcome function is not invariant under hyperbolic stabilization. □

For a nonsingular finite binary form, the Arf invariant controls the sign of the zero-set bias. This class-level statistic is exactly what the constant coordinate of Theorem 2.2 retains. Thus the compatible meaning of "recover Gold–Arf" is agreement of the Witt coordinate and outcome bias, not literal equality of all loaded boards.

5.3 Singular forms and cancellation

The nonsingular Witt group is cancellative. Arbitrary singular quadratic spaces form a different object: quasilinear directions already make their orthogonal-sum monoid noncancellative.

Theorem 5.3 (Quasilinear cancellation obstruction)**.** *Let F be a field of characteristic two. No isometry-invariant, orthogonal-sum-additive map from all finite-dimensional quadratic spaces over F to a cancellative commutative monoid can distinguish the restrictions of all forms to their polar radicals. In particular, no such family of impartial game values is faithful on arbitrary singular spaces.*

Proof. Fix $c \neq 0$ and write $L_c = \langle c \rangle$ and $Z = \langle 0 \rangle$. The invertible change of variables $(u,v) = (x+y, y)$ gives

$$L_c \perp L_c \cong L_c \perp Z, \qquad cx^2 + cy^2 = c(x+y)^2.$$

Let f be an additive isometry invariant with values in a cancellative commutative monoid. Then

$$f(L_c) + f(L_c) = f(L_c) + f(Z).$$

Cancellation gives $f(L_c) = f(Z)$. But the restrictions to the polar radicals are respectively the nonzero quasilinear form cx^2 and the zero form. Therefore f cannot retain the requested radical datum.

Finite impartial normal-play values are nimbers, and disjunctive sum is their xor addition, so their value monoid is cancellative. The conclusion applies coordinatewise to any additive impartial family. □

The obstruction leaves two coherent alternatives. One may retain orthogonal-sum additivity and classify only the cancellative quotient of the quasilinear monoid, or retain the full radical restriction in a separate nonadditive report. Requiring both is inconsistent.

6 Formal verification and external boundary

The Lean development has the focused import-only entry point `Ogdoad.Papers.WittRealization`, is also imported by `formal/Ogdoad.lean`, and contains no `sorry`, `admit`, or custom axiom declarations [1]. Its organization follows the proof rather than merely recording isolated lemmas.

The module `WittRealizationExact` proves the abstract exact-sequence algebra: the residue image is the transfer kernel, equal residues differ by a unique constant, and a constant detector gives the splitting $W \simeq C \times \ker(\text{transfer})$. The module `WittRealization` then defines the concrete presented coordinate space of Theorem 2.2, including the parity-zero ramified data. It constructs every constant, ramified, and trace-character observation; proves additivity, finite support, and faithfulness; realizes each bit by the literal empty-core $0/*$ arena; and proves the end-to-end equivalence

$$\text{all labelled P/N outcomes agree} \iff \text{the presented classes are equal.}$$

A single explicit classification equivalence transports this theorem to the actual quadratic Witt group. The same module constructs the one-place spike family and proves the fixed-finite-outcome obstruction at the coordinate level.

The arithmetic and boundary calculations are checked in four further modules. `WittTransfer` proves that every nonzero finite-field linear functional is trace against a nonzero square, verifies the simultaneous plane change, and proves the trace-dual Arf sum of Lemma 2.1. `WittRamification` proves termination, reachability, odd-pole normality, constant preservation, and triangularity of descending Artin–Schreier normalization. On top of it, `WittRamifiedRestriction` performs the finite unit-jet substitution, proves the sharp support bound, preserves its new constant term through normalization, and identifies the trace contribution $\eta_{e,u}$. Finally, `WittSingularBoundary` constructs the actual shear isometry $L_c \perp L_c \cong L_c \perp Z$ used in Theorem 5.3.

The remaining boundary is narrow and explicit. Mathlib has no characteristic-two quadratic Witt-group API, so Lean does not construct $W_q(\mathbb{F}_2(t))$ or prove the cited Aravire–Jacob exact sequence and local decomposition. Their identification with the concrete presented coordinate space is supplied through the one classification equivalence above. For the ramified theorem, Lean checks the complete finite substitution and normalization algorithm, but identifying the supplied unit jet with an actual power-series unit and transporting the unramified and ramified quadratic generators remain local-field mathematics in the paper. Likewise, the coordinate one-place spikes are checked internally, while their realization by the planes $[1, P^{-1}]$ and the infinitude of monic irreducibles are the standard global-field bridge used in the proof of the obstruction.

The Rust implementation computes the local triple (ϕ_0, ψ, ϕ_1) with canonical uniformizers for a supplied place. It does not yet expose the global coordinate theorem as a public Witt-class API or implement the ramified coordinate transport of Theorem 4.1.

7 Conclusion

The nonsingular realization problem has a complete classification answer. The unramified coordinate at infinity and all second residues give canonical coordinates for $W_q(\mathbb{F}_2(t))$. Finite-field trace characters present the wild odd-pole terms basis-freely, and the games 0 and $*$ realize the resulting finite-support bit vector exactly. Reciprocity is one parity equation on the ramified Arf bits. Under ramified scalar extension, the unit jet and finite inverse-Frobenius recurrence give the exact triangular coordinate law and its contravariant action on labels.

The stricter finite local-access request fails for a simpler reason: one fixed binary polar form supports infinitely many Witt-distinct place refinements, so no fixed finite P/N directory can be faithful. An infinite on-demand coordinate family, the sparse compiler constructed here, or a bounded place-and-conductor stratum is coherent, but the original conjunction is not. Beyond the nonsingular group, the quasilinear absorption relation separately rules out a faithful additive singular extension. Thus the coordinate theorem, its game realization, and its two limitations form one exact picture: the sparse nonsingular compiler is both faithful and as finite as the arithmetic permits.

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